Phylum Porifera

(Porous-pore; ferre-to bear)

Members of **Porifera**, commonly known as **sponges**, are among the **oldest living multicellular animals**. They originated in the **Precambrian era** and have survived successfully up to the present day.

About **8,550 species** of poriferans are known worldwide. Sponges reached their **greatest diversity during the Cretaceous period**. They play an important ecological role by contributing to **reef formation** and are a major source of **biogenic silica**, which forms flint deposits.

Sponges are **ubiquitous**, occurring in oceans across **all latitudes of the world**.
The scientific study of sponges is known as **Parazoology**.

General Features

- **Level of organisation:** Cellular level of organisation
- **Germ layers:** Diploblastic (functionally)
- **Body symmetry:** Mostly asymmetrical; some show radial symmetry
- **Coelom:** Absent (acoelomate)
- **Body plan:** Cell aggregate body plan
- **Digestion:** Intracellular digestion
- **Respiration:** Body surface respiration by diffusion
- **Circulation type:** Type-I (without blood)
- **Circulatory system:** Water canal system
- **Osmoregulation:** Osmoconformers
- **Excretion:** No excretory organs; wastes removed by diffusion
- **Mode of excretion:** Aminotelism (ammonia)
- **Skeleton:** Internal skeletal framework made of spicules or spongin
- **Nervous system:** Absent
- **Metamerism:** Absent
- **Fertilization:** Internal fertilization
- **Development:** Indirect development with larval stage
- **Body temperature:** Poikilothermic (ectothermic)

Characteristic Features

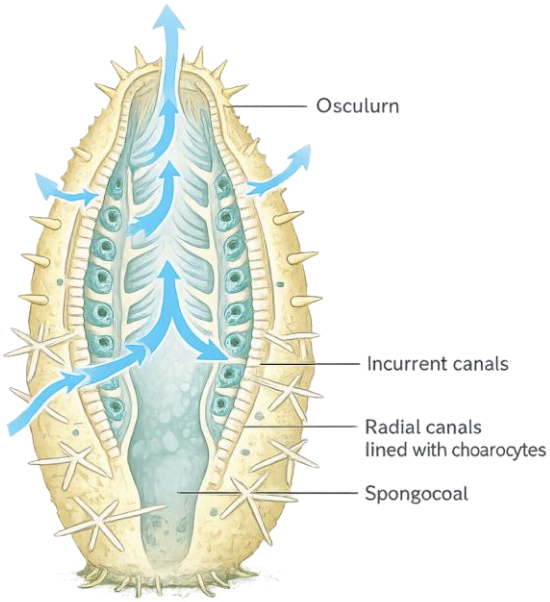
- Adult sponges are **sessile or sedentary organisms** and remain attached to an underwater object called the **substratum**.
- The body wall contains numerous small inhalant openings called **ostia**.
- These ostia open into a **single, central, and spacious cavity** called the **spongocoel** or **paragastric cavity**.
- Water leaves the body through **one or a few large exhalant openings** called the **osculum**.

Body Wall

- **Pinacoderm (dermal layer):**
  This is the **outer cellular layer**, made up of:
  - Flattened **pinacocytes**
  - Oval **porocytes**, which form the ostia
- **Choanoderm (gastral layer):**
  This is the **inner cellular layer**, composed of **choanocytes (collar cells)**.
  Choanocytes are the **characteristic cells of Porifera** and are responsible for:
  - Ingestion of food
  - Movement of water
  - Transfer of gametes
  - Differentiation of sex cells
- **Mesohyl layer:**
  This is a **non-cellular, jelly-like matrix** present between the pinacoderm and choanoderm.
  - It contains **spongin fibres and numerous spicules**
  - It also has many types of **amoebocytes**, derived from both pinacoderm and choanoderm
- The body wall is supported by a **skeleton** made of:
  - **Calcareous spicules** (calcium carbonate)
  - **Siliceous spicules** (silica)
  - **Spongin fibres**
  - Or a combination of these

Canal System (Water Canal System)

- The canal system, also called the **water canal system**, is a characteristic feature of all Porifera.
- Water enters the sponge body through **ostia**, passes into the **spongocoel**, and leaves through the **osculum**.
- This system helps in:
    - Food gathering
    - Respiration
    - Excretion (mainly ammonia)
    - Transfer of gametes by choanocytes
- The amount of water flowing through a sponge is very large and may reach **up to 20,000 times its body volume per day**.



Reproduction in Porifera

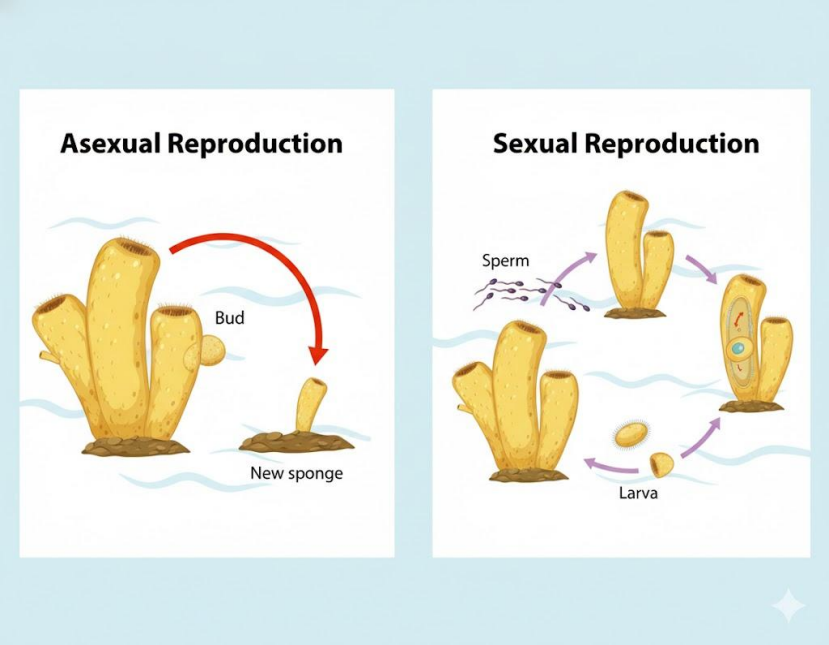
Sponges reproduce by **both asexual and sexual methods**, which helps them survive and spread in different environmental conditions.

Asexual Reproduction

- Asexual reproduction occurs by **fragmentation** and **budding**, which may be **external or internal**.
- **Internal buds**, produced during **unfavourable conditions**, are called **gemmules**.
- Sponges show a **high power of regeneration**, where even small body fragments can grow into a complete sponge.

Sexual Reproduction

- All sponges are **hermaphrodites**, meaning the sexes are not separate.
- Therefore, they are described as **monoecious** or **bisexual**.
- Sexual reproduction takes place by the formation of **haploid (n) male and female gametes**.
- Sperms are released into the water from one sponge and are carried by **water currents** into another sponge.
- Fertilisation occurs **inside the body** of the sponge, where the sperm fertilises the ovum **in situ**.
- Development includes **intermediate free-swimming larval stages**, such as **amphiblastula** and **parenchymula**, which help in the **dispersal of the species**.



Classification of Phylum Porifera

Based on the **nature of the skeleton**, phylum Porifera is divided into three main classes.

Class Calcarea

Calcarea, commonly known as **calcareous sponges**, are characterized by a skeleton made of **calcium carbonate spicules**.

- The spicules are composed of **calcium carbonate**.
- These sponges are **exclusively marine**.

- They are commonly found in **shallow coastal waters**, though some species occur at greater depths.
- About **400 species** of calcareous sponges have been described.
- Class Calcarea is divided into **three subclasses** based on the structure of their skeleton and spicules.

Class Hexactinellida

Hexactinellida, commonly known as **glass sponges**, are known for their delicate and intricate skeletons made of **silica (glass-like material)**.

- The skeleton consists of **six-rayed siliceous spicules**.
- These sponges are **exclusively marine**.
- They are mainly found in **deep-sea environments**, usually at depths of **200 metres or more**.
- Approximately **600–700 species** (around **680 documented species**) are known.
- The class Hexactinellida is divided into **two subclasses** based on skeletal and spicule structure.

Class Demospongiae

Demospongiae, commonly known as **demosponges**, form the **largest class** of sponges.

- The skeleton is made of **spongin fibres**, **siliceous spicules (not six-rayed)**, or both.
- This class includes **more than 90% of all sponge species**.
- Demosponges are found in a wide range of habitats:
    - Shallow marine waters
    - Deep-sea environments
    - **Freshwater ecosystems**
- The class is divided into **several subclasses**, based on skeletal structure, canal systems, and other morphological features.